Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 6: Physical and Chemical Changes

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: treating a single energetic/colour/irreversibility clue as decisive, confusing the solid product's mass with conservation of total mass, calling crystallisation or dissolving 'chemical', missing that one event can be physical and chemical at once, mistaking which condition a controlled rust experiment actually isolates, and reading galvanising's sacrificial action as mere physical sealing. Do not write on this paper — mark answers on your answer sheet.

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1. A teacher lists four claims about telling physical from chemical changes. Only one is always true. Which one?

(A) If a change gives out or takes in heat, it must be chemical, because energy only moves during reactions.

(B) If a change cannot be reversed, it must be chemical, because chemical changes are exactly the irreversible ones.

(C) If the original material can be recovered by a purely physical method, no new substance was formed, so the change was physical.

(D) If a colour change is seen, it must be chemical, because the substance has clearly turned into something else.

2. A 6 g strip of magnesium ribbon is burnt completely in open air and the white powder is collected and weighed. The powder weighs more than 6 g. The best explanation is that:

(A) burning created brand-new matter out of the flame's energy, adding mass to the metal.

(B) the dazzling light was absorbed by the powder and stored there as extra weight.

(C) magnesium combined with oxygen from the air, so the solid product (magnesium oxide) carries the original magnesium plus the added oxygen.
(D) mass can never change in any change, so the extra weight is just a weighing error.

3. Three changes are each irreversible by simple means: (i) a clay pot shattering, (ii) a boiled egg setting solid, (iii) a sheet of paper being shredded. Classify them correctly.

(A) all three are chemical, because none of them can be undone by a simple physical step.
(B) (i) physical, (ii) chemical, (iii) physical — only the egg forms a new substance; shattering and shredding leave the same material.
(C) all three are physical, because each only breaks an object into a different form.
(D) (i) chemical, (ii) physical, (iii) chemical — breaking and shredding rearrange bonds while the egg merely changes shape.

4. A hot saturated solution of potash alum is divided into two equal portions. Portion 1 is cooled slowly in a beaker; portion 2 is left in an open dish in the sun. Both finally give solid alum. Which statement is correct?

(A) Cooling is physical but evaporation is chemical, because heat from the sun changes the alum into a new substance.
(B) Both are chemical changes, because solid alum is a different substance from dissolved alum.
(C) Only evaporation gives true crystals; slow cooling can never separate a dissolved solid.
(D) Both are physical changes (crystallisation); slow cooling tends to give larger, purer crystals, while open evaporation drives off the water faster but can trap impurities.

5. Which list is correctly ordered as chemical, physical, chemical?

(A) Melting of butter; rusting of a nail; freezing of cream.
(B) Souring of milk; cooking of rice; dissolving of sugar.
(C) Crystallising sugar; burning of a candle; bending of a wire.
(D) Curdling of milk; sublimation of dry ice; ripening of a banana.

6. A clean iron plate is left in moist air. Each year a little brown rust forms and flakes off, and the bare plate underneath rusts again. After many years the plate is noticeably thinner and lighter, yet all the flakes collected weigh more than the metal that was lost. This is because:

(A) the rust flakes contain trapped air bubbles that make them seem heavier than the iron.
(B) rusting destroys iron and creates rust from nothing, so weights cannot be compared.

(C) the plate must have been impure, since pure iron would gain weight overall instead of thinning.

(D) rust is iron combined with oxygen and water, so the flakes are heavier than the iron consumed, even though the solid plate itself keeps losing iron.

7. Four sealed test tubes each hold an identical iron nail: (P) tap water with air above it, (Q) tap water with all air removed and the tube filled to the top, (R) dry air over a drying agent, (S) tap water with a layer of oil on top. After two weeks, which nail shows the most rust?

(A) P, because it is the only tube giving the nail both liquid water and a fresh supply of air (oxygen).

(B) R, because dry air is rich in oxygen and oxygen is the main thing rusting needs.

(C) Q, because being fully surrounded by water gives the nail the most moisture of all.

(D) S, because the oil layer keeps the nail warm and warmth speeds up rusting.

8. A car body is coated with zinc (galvanised). After a few years a deep scratch reaches the steel, yet the exposed steel at the scratch stays rust-free far longer than bare steel would. The chemistry behind this is best described as:

(A) zinc is more reactive than iron, so it corrodes in preference to the iron and goes on protecting the exposed metal even where the coating is broken.

(B) the zinc flows like a liquid into the scratch and physically seals it shut again.

(C) the scratch is so thin that air and moisture simply cannot fit into it to start rusting.

(D) the zinc has already turned the steel into stainless steel, which no longer rusts at all.

9. A student observes a change and says, 'Heat was given out, so it must be chemical.' Which single counter-observation most directly refutes the rule the student is using?

(A) Burning a strip of magnesium gave out heat and left a white powder that cannot be turned back.

(B) Stirring an anhydrous solid into water warmed the beaker, yet evaporating the water gave back the very same solid unchanged.

(C) Quicklime added to water gave out heat and formed a new substance, slaked lime.

(D) Wood burning in a stove gave out heat and left ash that is not the original wood.

10. A student blows exhaled air through a straw into clear limewater and it turns milky; on continuing to blow hard for a long time, the milkiness slowly clears again. The milky stage is caused by, and tells us:

(A) fine bubbles of breath scattering light, a purely physical cloudiness with no new substance.

(B) an insoluble white solid (calcium carbonate) forming from carbon dioxide in the breath, which is a chemical change because a new substance appeared.

(C) the water vapour in the breath condensing into a white mist inside the liquid.

(D) the limewater freezing into tiny ice crystals because the breath cooled it down.

11. A new bicycle chain comes coated in a thin film of oil. The shopkeeper says the oil 'stops the chain rusting.' How does the oil work, and what kind of protection is it?

(A) It chemically converts the iron surface into a rust-proof new metal, a chemical method.

(B) It forms a barrier that keeps air and moisture off the iron, a physical method of preventing the chemical change of rusting.

(C) It dries the iron out so completely that it can never react again, even if the oil wears off.

(D) It feeds extra oxygen to the iron in a controlled way so that no harmful rust can form.

12. Two clear solutions are mixed; the mixture stays clear at first, then turns cloudy and a fine solid settles out, and the liquid above is a different colour from either starting solution. The strongest single conclusion is that:

(A) a chemical change occurred, because a new insoluble solid (a precipitate) and a new colour both indicate new substances formed.

(B) a physical change occurred, because mixing two liquids is only a blending and can be undone by stirring.

(C) the solid is just dust that drifted in from the air and happened to settle while the colour faded.

(D) the solid is unreacted starting material that failed to dissolve and slowly fell to the bottom.

13. An iron nail is left for a day in blue copper sulphate solution. Afterwards: the nail is heavier and reddish-brown, and the liquid is pale green. Which set of statements is fully correct?

(A) The nail gained a coat of copper metal (so it is heavier), and the green liquid is iron sulphate; iron displaced copper because iron is more reactive.

(B) The nail rusted (the brown coat is rust) and the green colour is dissolved copper that leaked out of the solution.

(C) The nail lost weight because some iron dissolved away, and the green is the original copper sulphate fading in the light.

(D) Nothing chemical happened; iron simply dissolved like sugar and the colours are a trick of the lighting.

14. A change shows all of: a sharp new smell, bubbles of gas, AND the original material cannot be recovered by any simple physical step. A careful student still asks for the most decisive reason to call it chemical. The best answer is:

(A) the gas alone proves it, because only chemical changes can ever release a gas.

(B) the new smell alone proves it, because smells are never produced by physical changes.

(C) the inability to recover the original by a physical means shows a new substance formed, and the gas and new smell agree, so it is chemical.

(D) the three clues together cannot decide it; chemistry can only be confirmed by measuring how much heat is released.

15. To get pure crystals of common salt from rock salt that contains insoluble sand and clay, the correct order of steps is:

(A) heat the rock salt strongly until the sand and clay burn away, then collect the salt left behind.

(B) grind the rock salt finely and pick out the white salt grains from the darker dirt by hand.

(C) dissolve the rock salt in water, filter out the sand and clay, then evaporate or slowly cool the clear solution so salt crystals form.

(D) leave the rock salt in damp air until the sand rusts away, then scrape off the clean salt.

16. Which single pair correctly classifies BOTH members?

(A) Galvanising an iron sheet (the coating step) — chemical; cooking an egg — physical.

(B) Sublimation of naphthalene balls — chemical; rusting of a gate — physical.

(C) Dissolving washing soda in water — chemical; digesting a meal — physical.

(D) Fermentation of dough by yeast — chemical; sharpening a pencil with a blade — physical.

17. Near a burning candle, three things happen at once: solid wax melts to liquid, liquid wax rises up the wick, and wax vapour burns at the flame. The correct count of physical and chemical changes among these three events is:

(A) all three are chemical, because they all happen because of the flame's heat.

(B) all three are physical, because it is wax throughout until it finally disappears.

(C) two physical (melting and rising of the same wax) and one chemical (the wax vapour burning to new substances).

(D) one physical (melting) and two chemical (rising and burning), because moving up the wick changes the wax.

18. A factory must protect long iron pipes that will be buried in damp soil. They consider: (i) painting only, (ii) galvanising (zinc coating), (iii) wrapping in oily cloth. Which choice gives the most durable protection if the coating is likely to get scratched during burial, and why?

(A) Galvanising, because even at a scratch the more reactive zinc keeps corroding in place of the iron and protects the exposed metal.

(B) Painting, because paint chemically reacts with iron to make a permanent rust-proof layer that scratches cannot break.

(C) Oily cloth, because oil feeds the iron oxygen slowly so that only harmless rust forms.

(D) All three are identical in protection, since each merely seals the surface and a scratch ruins each one equally.

19. A student writes the rusting word equation as 'iron + air -> rust' and is told it is incomplete. The most precise correction within Class-7 scope is:

(A) iron + carbon dioxide + water -> rust, because the carbon dioxide in air is what attacks the iron.

(B) iron + nitrogen -> rust, because nitrogen is the most plentiful gas in the air.

(C) iron + oxygen (from air) + water -> hydrated iron oxide (rust); both oxygen and water are needed.

(D) iron + salt -> rust, because salt is the substance that combines with iron to make the brown solid.

20. After magnesium ribbon is burnt in air, the white ash is dropped into a little water and the water is tested; it now behaves like a mild base (turns red litmus blue). This shows that:

(A) the magnesium ribbon was simply melted and re-solidified, so it kept all its original properties.

(B) the litmus result is caused by leftover sandpaper grit from cleaning the ribbon, not by any new substance.

(C) burning produced a genuinely new substance whose properties differ from the metal, confirming a chemical change.

(D) magnesium metal itself turns litmus blue, so no new substance is needed to explain the result.

21. A solid block of camphor is heated gently in a dish covered with a cool inverted funnel; white camphor solid soon collects on the cool funnel, with no liquid seen. The two changes (camphor leaving the dish, and camphor reappearing on the funnel) are:

(A) both chemical, because heating broke the camphor down and then rebuilt a new white solid.

(B) both physical — sublimation (solid to vapour) then deposition (vapour to solid) of the same camphor.

(C) a chemical change (vaporising) followed by a physical change (re-forming the solid).

(D) a chemical change in which camphor reacts with air to make a new white powder on the funnel.

22. Which observation, taken entirely on its own, would be the WEAKEST single proof that a chemical change has occurred?

(A) A new gas with a sharp, unfamiliar smell was steadily given off.

(B) A fine white solid suddenly settled out when two clear liquids were mixed.

(C) The material burst into a flame, giving out bright light and strong heat.

(D) A solid was warmed and slowly turned into a clear liquid of the same material.

23. A teacher mixes lemon juice (an acid) with baking soda; the mixture fizzes and the bubbles, when passed into limewater, turn it milky. From this, the correct conclusions are:

(A) a chemical change occurred and the gas given off is carbon dioxide, because it turns limewater milky.

(B) a physical change occurred, because fizzing is just dissolved air escaping like in a cold drink.

(C) the gas is oxygen, because the bubbling is vigorous and oxygen supports such activity.

(D) the gas is hydrogen, because acids always release hydrogen and hydrogen forms bubbles.

24. Stainless-steel cutlery and a chrome-plated tap both resist rust for years. A student asks why iron in each does not rust. The best shared explanation is:

(A) in both, the iron has been chemically turned into pure chromium, which cannot rust.

(B) both work only because the items are always kept dry and never touch water.

(C) in both, a rust-resistant layer or alloy containing chromium keeps a protective surface between the iron and the air and moisture.

(D) both speed up the iron's reaction with oxygen so fast that the iron is used up before it can rust.

25. A student claims, 'Cooking is always a chemical change.' Which everyday kitchen step shows the claim is too broad and is actually physical?

(A) Boiling an egg until the white turns solid and opaque.

(B) Toasting bread until the surface browns and gives a roasted smell.

(C) Melting butter in a warm pan, where it later re-solidifies on cooling as the same butter.

(D) Caramelising sugar in a hot pan until it turns brown and bitter.

26. An experiment keeps four iron nails identical except for one factor each: nail 1 in moist air with oxygen, nail 2 in moist air with the oxygen removed, nail 3 in dry air with oxygen, nail 4 in dry air with oxygen removed. Nail 1 rusts heavily; the other three barely rust. The clearest single conclusion is:

(A) rusting needs BOTH oxygen and moisture together, since rust appears only when both are present.

(B) rusting needs only oxygen, since the nail in dry air with oxygen should have rusted just as much.

(C) rusting needs only moisture, since the nail in moist air without oxygen should have rusted just as much.

(D) rusting needs neither oxygen nor moisture, since iron rusts in time no matter what.

27. Inside an unopened tin of food, the iron container is coated with a layer of tin. A child argues, 'Tin and zinc protect iron in exactly the same way.' Within Class-7 scope, the most accurate reply is:

(A) they are identical, because both turn the iron into a brand-new rust-proof metal underneath.

(B) tin makes the iron rust faster, while zinc stops it, so they work in completely opposite ways.

(C) both coat the iron to keep air and moisture away, but if the coating is scratched the protection is weakened, so the simplest shared idea is that each forms a protective surface layer over the iron.

(D) neither tin nor zinc has anything to do with rusting; the food itself keeps the iron from rusting.

28. Which comparison of rusting, burning, and respiration is correct?

(A) Only burning is chemical; rusting and respiration are physical because the iron and the body stay the same.

(B) All three are physical, because none of them makes any new substance.

(C) Rusting and burning are chemical, but respiration is physical because it only releases warmth.

(D) All three are chemical changes in which a substance combines with or uses oxygen; burning is fast, respiration is a slow controlled use inside the body, and rusting is slow surface oxidation.

29. Pure salt is dissolved in water; half is left in the sun to dry and gives back white salt, while the other half is heated very strongly in a metal dish and also leaves white solid. A student says the strongly heated half 'must be a new substance because so much heat was used.' The correct verdict is:

(A) the strongly heated half is chemically new, because strong heat always creates a new substance.

(B) both halves give back the same salt; using a lot of heat to evaporate water is still a physical change, since no new substance forms.

(C) the sun-dried half is physical but the strongly heated half is chemical, because the amount of heat decides the type.

(D) both halves are chemical, because evaporating water from salt always makes a fresh compound.

30. Which change is the odd one out — the only PHYSICAL change in the list?

(A) Bread dough rising and baking into a loaf.

(B) A cut apple turning brown after a while in the air.

(C) Green grapes drying into raisins by losing water and turning brown and sweet-sour.

(D) Wax solidifying as it drips down the side of a candle and cools.

31. A pale-green solution of iron sulphate is left in an open dish. Slowly some of it turns brownish at the surface. Within Class-7 ideas, the most reasonable explanation is that:

(A) the iron in the solution is reacting with oxygen and moisture from the open air, a chemical change that forms a rust-like brown substance.

(B) the green colour is simply fading in the light, a physical change with no new substance.

(C) the solution is crystallising, and the crystals just happen to look brown.

(D) the iron sulphate is dissolving further, which always turns solutions brown.

32. Which paired classification is fully correct?

(A) Burning of a candle wick — physical; melting of candle wax — chemical.

(B) Crystallisation of sugar from syrup — chemical; cutting of a wire — chemical.

(C) Digestion of food — physical; ripening of a mango — physical.

(D) Souring of milk into curd — chemical; condensation of steam into water droplets — physical.

33. A student insists, 'Any change that needs heat to happen is a chemical change.' Which single example best refutes this rule?

(A) Heating sugar until it caramelises into a brown, bitter substance.

(B) Heating limestone strongly until it gives off a gas and leaves a new solid.

(C) Heating ice until it melts to water, which freezes back to the same ice on cooling.

(D) Heating magnesium ribbon until it burns to a white powder.

34. A bottle of fresh limewater is left loosely corked for a few weeks and a faint white skin forms on its surface. The most likely reason within Class-7 chemistry is:

(A) the water slowly evaporated, leaving a physical film of dust on top.

(B) carbon dioxide from the air slowly reacted with the limewater to form a thin layer of insoluble white solid, a chemical change.

(C) the limewater froze a little at the surface where it was coolest.

(D) oxygen from the air dissolved in and bleached the surface white.

35. When green leaves photosynthesise, carbon dioxide and water are turned into glucose and oxygen using sunlight. A student calls it physical 'because the leaf just gets greener.' The correct reasoning that makes it chemical is:

(A) sunlight is absorbed, and absorbing light is always a chemical change.

(B) new substances (glucose and oxygen) with properties unlike the starting materials are produced, which is the test for a chemical change.

(C) water changes from liquid to vapour inside the leaf, and changing state is chemical.

(D) the process can be reversed by simply placing the plant in the dark, and reversible changes are chemical.

36. A bar magnet is stroked along a steel needle, and the needle becomes a magnet that can pick up pins. Later the needle is heated strongly and dropped, and it loses its magnetism. Throughout, the needle stays steel. Magnetising and de-magnetising the needle are:

(A) both chemical changes, because the needle gains and then loses magnetic energy.

(B) a chemical change (magnetising) followed by a physical change (de-magnetising).

(C) rusting, because heating steel in air always corrodes it into a new substance.

(D) both physical changes, because the needle remains the same steel and no new substance is formed.

37. A spoon of common salt is added to a cup of plain water and to a cup of water that already has copper sulphate dissolved in it. In the plain water the salt just dissolves; nothing else is seen. In the copper sulphate water it also dissolves and nothing new appears. The correct conclusion is:

(A) in the copper sulphate cup a chemical reaction occurred, because two dissolved substances always react.

(B) in both cups the salt only dissolved (a physical change); dissolving salt does not make a new substance, so no reaction occurred.

(C) the salt was destroyed in both cups, because dissolved salt can never be recovered.

(D) dissolving in copper sulphate water is chemical while dissolving in plain water is physical, because the colour was different.

38. A long iron railing rusts noticeably faster near the seashore than the same railing far inland. Within Class-7 ideas, the best reason is:

(A) the sea air carries more moisture and dissolved salt, and salt speeds up rusting, so both moisture and salt accelerate the corrosion.

(B) sea air is hotter, and heat alone is what turns iron into rust.

(C) the sea air contains extra carbon dioxide, which is the gas that rusts iron.

(D) the railing near the sea is made of a weaker iron that rusts on its own without any air or moisture.

39. A change releases heat AND forms a new substance. A second change absorbs heat AND forms a new substance. A third change releases heat but forms NO new substance. How many of these three are chemical changes?

(A) Two, because forming a new substance defines a chemical change regardless of whether heat is taken in or given out.

(B) One, because only a change that releases heat can be chemical.

(C) Three, because any change involving heat is chemical.

(D) Zero, because heat involvement always means the change is only physical.

40. A blacksmith heats an iron rod till it is red-hot, hammers it into a curved hook, and quenches it in water; ignoring the thin surface scale, the rod is still iron afterwards. Two events are: (i) shaping the red-hot iron, and (ii) the iron glowing red. These are best described as:

(A) both chemical, because heating iron until it glows always turns it into a new substance.

(B) both physical — shaping changes only the form, and glowing is hot iron giving off light, with the iron staying iron in both.

(C) a physical change (shaping) and a chemical change (glowing), because the glow proves a reaction.

(D) rusting and crystallisation, because hot iron quenched in water rusts and forms crystals.

41. Which statement comparing crystallisation and rusting is correct?

(A) Crystallisation is a physical change that gives back the same substance as pure crystals, while rusting is a chemical change that forms a new substance from iron, oxygen and water.

(B) Both are chemical changes, because both produce a solid that was not there before.

(C) Both are physical changes, because in each the metal or salt stays the same substance.

(D) Crystallisation is chemical (new crystals form) while rusting is physical (iron stays iron under the rust).

42. To test whether OXYGEN (and not just any gas) is needed for rusting, while keeping water present, the best pair of set-ups is:

(A) one nail in dry air with a drying agent versus one nail in moist air — to compare two kinds of air.

(B) one nail in water open to ordinary air versus one nail in water that has been boiled to expel air and then sealed under oil — both wet, differing only in whether air (oxygen) can reach the water.

(C) one nail in salty water versus one nail in plain water — both open to air, to compare two waters.

(D) one nail painted versus one nail left bare in damp air — to compare a sealed nail with an open one.

43. A coloured fruit drink concentrate is bright red. When water is added it becomes pale pink; when more water is added it becomes nearly colourless. A student says 'the colour change proves a chemical reaction with water.' The correct verdict is:

(A) a chemical change occurred, because any colour change is proof of a new substance.

(B) no chemical change occurred; diluting only spreads the same colouring substance through more water, a physical change recoverable by evaporation.

(C) a chemical change occurred, because adding water always reacts with the dye to weaken it.

(D) it is impossible to decide, because colour changes can never be classified as physical or chemical.

44. A boiled egg is irreversible; a smashed egg shell is also irreversible. A student concludes 'both are chemical because both are irreversible.' The verdict for the cooking is right but the reasoning is wrong because:

(A) irreversibility always guarantees a chemical change, so the reasoning is actually fine.

(B) both the cooking and the smashing are in fact physical, so the verdict is wrong.

(C) neither cooking nor smashing forms a new substance, so neither is chemical.

(D) irreversibility alone does not prove a chemical change; cooking is chemical because new substances form, while a smashed shell is irreversible yet still the same shell material (physical).

45. A change is studied with four tests: (1) tried to recover the starting material physically — could not; (2) measured temperature — it rose; (3) looked for a colour change — none seen; (4) checked for a gas — none seen. Which single result is the strongest evidence that the change is chemical?

(A) Test 1: the starting material could not be recovered by a physical method, indicating a new substance formed.

(B) Test 2: the temperature rose, which by itself proves a chemical change.

(C) Test 3: no colour change was seen, which proves the change must be chemical.

(D) Test 4: no gas was seen, which proves a new substance formed.

46. Two changes are compared: (i) iced water left out warms up and the ice melts; (ii) a slice of bread left out goes mouldy and smells bad. Choose the correct classification and the reason that distinguishes them.

(A) both physical, because in each something is simply left out in the open air to change on its own.

(B) both chemical, because both changes happen because of warmth from the surroundings.

(C) (i) physical, because the ice only changes state to the same water; (ii) chemical, because mould growth produces new substances and a new smell.

(D) (i) chemical because heat was absorbed, and (ii) physical because the bread merely changed appearance.

47. A coil of bright copper wire is heated strongly in a flame and turns black on the surface; when scraped, fresh copper shows beneath. The black coating is best explained as:

(A) soot from the flame settling on the wire, which is only physically stuck on.

(B) a new substance (a copper-oxygen compound) formed where copper combined with oxygen in the air — a chemical change at the surface.

(C) the copper merely getting hot, so the black is just the colour of hot copper.

(D) the copper melting and re-freezing black, a physical change of state.

48. Which statement about the energy changes accompanying physical and chemical changes is correct?

(A) Only chemical changes can release energy; physical changes never do.

(B) Only physical changes can absorb energy; chemical changes always release it.

(C) All energy-releasing changes are chemical and all energy-absorbing changes are physical.

(D) Both physical and chemical changes can absorb or release energy; energy flow alone does not tell you which type a change is.

49. A glass of warm sugar solution is left uncovered for several days; the water evaporates and clear sugar crystals appear on the glass, with a few black specks where some sugar near the dry edge browned. The crystal-forming and the browning are, respectively:

(A) a physical change (crystallisation of the same sugar) and a chemical change (sugar browning into a new substance).

(B) both physical changes, because it is all sugar coming out of the same solution.

(C) both chemical changes, because the sugar separated from the water in each case.

(D) a chemical change (crystallising) and a physical change (browning), because new crystals are a new substance.

50. A student melts a small lump of solder (a metal that melts easily) on a tin lid, lets it cool to a hard lump, then re-melts it. Each time it is the same shiny metal. A friend says 'melting a metal must be chemical because metals are special.' The correct response is:

(A) yes, melting a metal is chemical, because the heat changes the metal into a new substance each time.

(B) melting and re-solidifying the solder are physical changes of state; the metal keeps its identity, so being a metal does not make melting chemical.

(C) melting is physical but cooling is chemical, because the cooled lump is a new solid.

(D) it is rusting, because heating a metal in air on a lid always corrodes it into a new substance.

51. Which one of these correctly identifies BOTH a physical change AND a chemical change occurring during the SAME everyday event?

(A) Ice melting in a drink: the ice melting is physical and the drink getting colder is chemical.

(B) Salt dissolving in water: the salt spreading out is physical and the water turning salty is chemical.

(C) A nail being bent: the bending is physical and the nail getting warm from bending is chemical.

(D) A candle burning: the solid wax melting is physical, while the wax vapour burning at the flame is chemical.

52. A laboratory note reads: 'On mixing, the beaker grew warm, a fine yellow solid settled out, a faint new smell arose, and a gas bubbled off; nothing could be recovered by simply evaporating.' How many of these five observations are recognised clues of a chemical change?

(A) Three, because warmth and the failure to recover by evaporation are not chemical-change clues.

(B) Two, counting only the gas and the precipitate, since smells and heat are unreliable.

(C) Four, because warmth alone is never a clue to a chemical change.

(D) Five — warmth, a precipitate, a new smell, a gas evolved, and failure to recover the original by a physical step are all clues that a new substance formed.

53. A student is told that crystals of copper sulphate grown by slow cooling are 'pure' while the impure starting powder was dirty. The reason crystallisation purifies is best stated as:

(A) the impurities chemically react with the copper sulphate and are turned into more copper sulphate.

(B) the heat used to dissolve the powder burns all the impurities away completely.

(C) the crystals are a brand-new compound that no longer contains any of the original impurities.

(D) the soluble copper sulphate dissolves and re-forms as crystals while the insoluble impurities are filtered off and any soluble impurities stay behind in the leftover solution.

54. Four iron objects are stored for a year: (W) a tool wiped with grease and kept indoors, (X) a gate painted and kept outdoors, (Y) a chain left bare in a humid shed, (Z) a knife coated with zinc and kept outdoors. Rank them from LEAST to MOST rusted, assuming all coatings stay intact.

(A) W and Z least, X next, Y most — because intact grease, zinc, and paint all block or sacrificially protect the iron, while the bare chain in humid air rusts freely.

(B) Y least, then X, then Z, then W most, because grease attracts moisture and speeds rusting fastest.

(C) all four rust equally, because iron always rusts in a year no matter how it is stored.

(D) Z most rusted, because the zinc coating reacts with air and rusts faster than bare iron.

55. Two students debate: Aman says 'a change is chemical only if you can see something happen, like a colour, gas, or flame.' Bina says 'some chemical changes show no obvious clue at the moment, yet a new substance has formed.' Who is correct, and why?

(A) Aman, because if no clue is visible then no new substance can possibly have formed.

(B) Bina, because the real test is whether a new substance formed, which can be true even when no dramatic clue is seen at the time.

(C) Aman, because chemical changes are defined as the changes that produce a visible signal.

(D) neither, because a change with no visible clue is always physical by definition.

56. A piece of bright iron wire is held in a flame and burns with orange sparks, leaving dark crumbly bits. A piece of the same wire left in damp air for weeks slowly turns brown and flaky. Both events are best described as:

(A) both chemical changes in which iron combines with oxygen — fast and fiery in the flame, slow as rusting in damp air.

(B) the flame event is chemical (burning) but the slow browning is physical, because slow changes are physical.

(C) both physical, because the wire is still made of iron in both cases.

(D) the slow browning is chemical (rusting) but the fiery burning is physical, because sparks are just hot bits flying off.

57. A student writes: 'Because dissolving sugar and crystallising it back are both reversible, they prove that reversible changes are physical.' Which case best supports the claim that reversibility goes with physical changes?

(A) Burning a candle and then trying to collect the smoke back into wax.

(B) Cooking an egg and then cooling it to get the runny egg back.

(C) Dissolving salt in water and then evaporating to recover the same salt — reversible, with the same substance throughout.

(D) Rusting an iron nail and then drying it to turn the rust back into shiny iron.

58. An iron gate is painted, but after a year a small chip in the paint shows a spreading rust spot that lifts the surrounding paint. The best explanation is:

(A) the paint itself slowly turned into rust across the whole gate at once.

(B) the chip let in extra paint solvent, which is what corroded the iron.

(C) where the paint chipped, air and moisture reached the bare iron and rust formed; the expanding rust pushed off nearby paint, exposing more iron to rust further.

(D) the rust spot proves paint speeds up rusting compared with leaving iron bare.

59. A controlled experiment finds: nail in dry air — no rust; nail in pure boiled water sealed from air — no rust; nail in ordinary water open to air — rusts. A student concludes 'water alone causes rust.' What is the correct conclusion?

(A) Water alone causes rust, since the nail in open water rusted.

(B) Air alone causes rust, since the dry-air nail was exposed to plenty of air.

(C) Water alone is not enough; rusting needs both water and air (oxygen), since the sealed boiled-water nail had water but no air and did not rust.

(D) Neither water nor air is needed; the open-water nail rusted only because it was disturbed.

60. A student summarises four ideas. Which single idea is FALSE within Class-7 science?

(A) A chemical change forms one or more new substances, while a physical change does not.

(B) Some physical changes, such as freezing, release heat to the surroundings.

(C) Crystallisation gives pure crystals of the same substance and is a physical change.

(D) Every change that produces a gas is a chemical change.

Solutions — Science (Class 7), Chapter 6: Physical and Chemical Changes — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	B	31	A	41	A	51	D
2	C	12	A	22	D	32	D	42	B	52	D
3	B	13	A	23	A	33	C	43	B	53	D
4	D	14	C	24	C	34	B	44	D	54	A
5	D	15	C	25	C	35	B	45	A	55	B
6	D	16	D	26	A	36	D	46	C	56	A
7	A	17	C	27	C	37	B	47	B	57	C
8	A	18	A	28	D	38	A	48	D	58	C
9	B	19	C	29	B	39	A	49	A	59	C
10	B	20	C	30	D	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The single reliable rule is the new-substance test, and recovering the starting material by a physical means (such as evaporating to get dissolved salt back) proves no new substance ever formed, so the change was physical. The 'heat means chemical' idea fails because dissolving some solids or even melting and freezing exchange heat while staying physical. The 'irreversible means chemical' idea fails because smashing glass is irreversible yet physical. 'Colour change means chemical' fails because heating a wire to glow, or merely diluting a coloured liquid, changes colour with no new substance. Only the recoverability criterion holds in every case.

2. (C) Burning magnesium is a chemical change in which the metal combines with oxygen to make magnesium oxide. The solid product therefore contains the original magnesium atoms plus oxygen atoms taken from the air, so the solid alone weighs more than the ribbon did. Energy (the light and heat) is not matter and does not add weighable mass, so 'matter created from energy' and 'light stored as weight' are both wrong. Total mass counting the oxygen taken from the air is conserved, but that does not stop the solid by itself from gaining mass, so 'mass can never change' misreads conservation as applying to the solid alone.

3. (B) Irreversibility alone does not decide the type; only the formation of a new substance does. The shattered pot is still the same clay material in pieces (physical), and shredded paper is still paper (physical), even though neither can be reassembled. Cooking the egg forms new substances from the egg proteins and cannot be undone (chemical). So the pattern is physical, chemical, physical. 'All chemical because irreversible' and 'all physical because only the form changes' both apply one feature blindly; the bond-rearranging description is exactly backwards, since it is the egg, not the pot or paper, where new substances form.

4. (D) Crystallisation is a physical change: the dissolved alum simply comes out of solution as crystals of the very same substance, whether the solution is cooled (which lowers how much it can hold) or evaporated (which removes the water). Slow cooling usually grows fewer, larger, cleaner crystals, while fast open evaporation can leave impurities behind in the solid. Dissolved alum and solid alum are the same compound, so it is not chemical, and the sun's gentle heat does not transform alum into anything new. Both routes yield real crystals, so 'only evaporation works' is false.

5. (D) Curdling forms new acidic substances (chemical); dry ice subliming is solid carbon dioxide turning to gas of the same substance (physical); a banana ripening develops new sugars and aroma compounds (chemical) — so chemical, physical, chemical. 'Melting butter, rusting, freezing cream' is physical, chemical, physical. 'Souring, cooking, dissolving' is chemical, chemical, physical. 'Crystallising sugar, burning candle, bending wire' is physical, chemical, physical. Only the first list matches the required order.

6. (D) Rusting combines iron with oxygen and water to form hydrated iron oxide, so each gram of iron that reacts ends up in a flake that also contains added oxygen and water and therefore weighs more than the iron alone. That is why the collected flakes outweigh the metal lost, while the plate itself thins because iron keeps leaving its surface as rust that crumbles away. Trapped air is far too light to explain the gain; the mass comes from chemically combined oxygen and water. Rust is not made from nothing. Even pure iron thins as a plate while its flakes gain mass, so impurity is not the reason.

7. (A) Rusting needs iron, oxygen, and water together. Tube P has both liquid water and air above it, so it rusts most. Tube R has air but no moisture, so it barely rusts. Tube Q has plenty of water but the air was removed, so it lacks oxygen and rusts very little. Tube S has water but the oil seals out air, so again little oxygen reaches the nail. Dry air alone cannot rust iron because moisture is missing, and merely being surrounded by water without dissolved air does not help, while oil seals air out rather than warming the nail.

8. (A) Galvanising protects partly as a barrier but crucially because zinc is more reactive than iron: at the scratch the surrounding zinc corrodes first and keeps shielding the bare steel, which is why galvanised steel resists rust even where the coating is broken. Solid zinc does not flow to re-seal a scratch. A scratch easily admits air and moisture, so size is not the reason. Galvanising lays a zinc coat over the steel; it does not convert the steel into stainless steel, which would require chromium mixed throughout the metal.

9. (B) The rule 'heat given out means chemical' is broken by a physical change that still gives out heat. Dissolving an anhydrous solid can warm the water, yet evaporating recovers the same solid, proving no new substance formed — a physical change that released heat. Burning magnesium and burning wood do give out heat, but they are chemical (new substances form), so they support the rule rather than refute it. Quicklime plus water also forms a new substance with heat, again chemical. Only the dissolving-and-recovering example shows heat without any new substance, defeating the rule.

10. (B) Exhaled air is rich in carbon dioxide, which reacts with limewater to form insoluble calcium carbonate; its fine white particles make the liquid look milky, and the appearance of a new insoluble solid is a clear chemical-change clue. The clouding is a chemical product, not light-scattering by gas bubbles, which would not persist. Condensing breath vapour cannot form a white solid inside the liquid. The breath does not freeze the limewater; the milkiness appears the moment the carbon dioxide reacts, not from cooling.

11. (B) Oil and grease protect by coating the iron so that air and moisture cannot reach it; the oil itself does not change the iron, so it is a physical barrier method that prevents the chemical change of rusting. It does not turn the iron into a new metal — that would be a chemical change such as galvanising. It does not permanently dry the metal; if the film wears away, air and moisture can reach the iron and rusting begins. Adding oxygen would speed rusting, not prevent it, since rusting needs oxygen.

12. (A) A precipitate forming where the mixture was clear, together with a new colour in the liquid, are two independent clues that new substances were produced — a chemical change. Mixing solutions is not always physical: here a reaction makes an insoluble product, which simple stirring cannot undo. Stray dust would not appear throughout the liquid at the moment of mixing nor change the colour. The solid is not undissolved starting material, because both solutions were clear before mixing; the solid is a product of the reaction between them.

13. (A) Iron is more reactive than copper, so it displaces copper from copper sulphate: copper metal deposits on the nail as a reddish-brown coat, making the nail heavier, while iron goes into solution as pale-green iron sulphate. The brown coat is copper, not rust, since rust needs air and water and is a different oxide. The nail gains mass from the copper coat, so 'lost weight' is wrong, and the green is a genuine new substance, not light playing tricks or the blue merely fading. Iron does not dissolve unchanged like sugar; a displacement reaction occurs.

14. (C) The decisive criterion is whether a new substance formed; failure to recover the starting material by a physical method shows one did, and the new smell and evolved gas are confirming clues of new substances. A gas can also be released physically (boiling water gives steam, opening a fizzy drink lets dissolved gas escape), so a gas alone is not proof. A new smell usually signals new substances but is not by itself the formal test. Measuring heat does not settle it, since physical changes also exchange heat; the recovery test is the reliable one.

15. (C) Crystallisation purifies a soluble solid by dissolving it (the salt goes into solution while sand and clay do not), filtering to remove the insoluble dirt, then evaporating or cooling so pure salt crystals separate from the clear solution. Strong heating does not burn away sand and clay, which are not combustible, and would not separate them from salt. Hand-picking cannot reliably separate finely mixed material or give pure crystals. Sand and clay do not rust — rusting is specific to iron — so damp air would not clean the salt.

16. (D) Yeast fermenting dough produces new substances such as carbon dioxide and alcohol, so it is chemical, while sharpening a pencil only removes shavings, leaving the same wood and graphite, so it is physical — both labelled correctly. Galvanising's coating step lays zinc over iron without changing the iron, so it is physical, and cooking an egg forms new substances, so it is chemical — that pair is reversed. Naphthalene subliming is a state change (physical) and rusting forms a new oxide (chemical) — reversed. Dissolving washing soda is physical (the solid is recoverable) and digestion is chemical — reversed too.

17. (C) Melting wax and liquid wax rising up the wick both keep the same wax — these are physical changes (a state change and a movement). Only at the flame does wax vapour combine with oxygen to form new substances such as carbon dioxide, water and soot — a chemical change. So the count is two physical and one chemical. Being caused by heat does not make a change chemical (melting is heat-driven yet physical), so 'all chemical' is wrong. The wax does change chemically at the flame, so 'all physical' is wrong, and merely rising up the wick forms no new substance, so it is not chemical.

18. (A) Galvanising is best for a coating likely to be scratched because zinc is more reactive than iron; at a scratch the zinc corrodes preferentially and goes on protecting the exposed iron, unlike a pure barrier. Paint is only a barrier, not a chemical bond with the iron, and a scratch lets the bare iron rust. Oil also only seals; it does not feed oxygen, and feeding oxygen would speed rusting anyway. The three are not equal: paint and oil fail at a scratch, while galvanising's sacrificial zinc keeps working, so a scratch does not ruin all three equally.

19. (C) Rusting requires iron to combine with oxygen from the air and with water (moisture) to form hydrated iron oxide, so 'iron + air' is incomplete because it leaves out water and does not specify oxygen as the reactive gas. Carbon dioxide is not the reactant in rusting — that gas is used in the limewater test, not corrosion. Nitrogen, though most plentiful in air, does not rust iron. Salt can speed up rusting but is not consumed into the rust and cannot replace the essential oxygen and water, so 'iron + salt' is wrong.

20. (C) The white ash dissolving to give a mildly basic solution shows it has properties the shiny metal ribbon never had, which is direct evidence that burning made a new substance (magnesium oxide) — a chemical change. The ribbon was not merely melted and re-solidified; it was chemically converted, so it does not keep the metal's properties. The basic behaviour comes from the oxide product, not from sandpaper grit used to clean the ribbon. Magnesium metal does not turn litmus blue by itself, so a new substance is indeed needed to explain the result.

21. (B) Camphor sublimes directly from solid to vapour when warmed, and the vapour deposits back to solid camphor on the cool surface — both are physical state changes of the very same substance, with no new substance formed (no liquid stage is needed). Heating does not break camphor into new substances and rebuild it, so 'both chemical' is wrong. Vaporising is not chemical; the substance stays camphor, so the mixed labels are wrong. Camphor does not react with air to make a new powder; the white solid on the funnel is the original camphor.

22. (D) A solid melting into a liquid of the same material is simply a change of state and is almost always physical, so by itself it is the weakest evidence of chemistry. A new-smelling gas suggests a new substance, a precipitate from mixing two solutions is a recognised chemical clue, and bursting into flame with light and heat points to combustion — all are far stronger indicators of a chemical change than a mere change of state.

23. (A) The acid reacting with baking soda makes new substances and releases a gas that turns limewater milky; only carbon dioxide gives that milky test, so the gas is carbon dioxide and the change is chemical. The fizzing is not dissolved air escaping as in a fizzy drink (which is physical); here a reaction generates new gas. The gas is not oxygen — oxygen does not turn limewater milky. Within this Class-7 limewater test the milkiness identifies carbon dioxide, not hydrogen, so 'hydrogen' is wrong.

24. (C) Chrome plating coats the iron with a thin rust-resistant chromium layer, and stainless steel is an iron alloy containing chromium that gives a corrosion-resistant surface — in both, a chromium-bearing surface keeps air and moisture from attacking the iron beneath. The iron is not transmuted into pure chromium; it stays iron with a protective surface. They resist rust even when wet (cutlery is washed daily), so 'only if kept dry' is wrong. They slow the iron's reaction with oxygen rather than speeding it up, so the last option is the opposite of the truth.

25. (C) Melting butter only changes its state, and it re-solidifies as the same butter on cooling, so it is physical — showing not every kitchen step is chemical. Boiling an egg sets new substances from the proteins (chemical), toasting bread browns it by forming new substances with a roasted smell (chemical), and caramelising sugar forms new brown, bitter substances (chemical). All three of those are genuine chemical changes, so only melting butter refutes the over-broad claim.

26. (A) Only nail 1, which has both oxygen and moisture, rusts heavily; removing either oxygen (nail 2) or moisture (nail 3), or both (nail 4), almost stops rusting. That pattern shows both factors are needed together. 'Only oxygen' is wrong because nail 3 had oxygen in dry air yet barely rusted. 'Only moisture' is wrong because nail 2 had moisture without oxygen yet barely rusted. 'Neither' is wrong because nail 1 rusted heavily precisely when both were present.

27. (C) Both tin and zinc are applied as protective surface coatings that keep air and moisture off the iron, which is the core Class-7 idea about coating metals to prevent rusting; in both cases an intact coat protects and a break in the coat lets attack begin. Neither turns the iron into a new metal beneath; the iron stays iron. Tin does not make iron rust faster — it is a protective coat, not an accelerator. The protection comes from the metal coatings, not from the food, so the last option is wrong.

28. (D) Rusting, burning, and respiration all involve oxygen and all form new substances: burning oxidises a fuel rapidly with heat and light, respiration uses oxygen to break down sugar inside the body releasing energy, and rusting is slow oxidation of iron at its surface — so all three are chemical. They are not physical: each makes new substances. Respiration is not merely physical warmth; it breaks sugar into carbon dioxide and water, so the options calling some of them physical are wrong.

29. (B) Evaporating the water — whether slowly in the sun or quickly with strong heat — only removes the water and leaves the same salt behind, so both are physical changes; the amount of heat used does not change a physical separation into a chemical one. Strong heat does not automatically create a new substance; here it merely drives off water faster. The salt is recoverable and unchanged in both halves, so neither half is chemical, and evaporating water from salt does not make a new compound.

30. (D) Liquid candle wax cooling and solidifying is just a change of state of the same wax, so it is physical and the odd one out. Dough baking into bread forms new substances (chemical). A cut apple browning in air is a chemical change as it reacts with oxygen, forming new brown substances. Grapes drying into raisins lose water but also develop new substances as they age and change flavour and colour, which is treated as a chemical change here rather than mere drying; the safest single physical example is the solidifying wax.

31. (A) Iron sulphate exposed to open air can react with oxygen and moisture to form a brownish iron-oxide-type substance at the surface — a chemical change producing a new substance, similar to the chemistry of rusting. The green is not merely fading in light; a new brown substance appears. Crystallisation would give green crystals of the same iron sulphate, not a brown surface layer. Dissolving further would not turn the solution brown; the brown colour signals a new substance from reaction with air.

32. (D) Milk souring into curd forms new acidic substances (chemical), and steam condensing into water droplets is a change of state of the same water (physical) — both labelled correctly. A burning wick makes new substances (chemical) and melting wax is a state change (physical), so that pair is reversed. Crystallising sugar is physical and cutting a wire is physical, so calling them chemical is wrong. Digestion is chemical and mango ripening is chemical, so calling them physical is wrong.

33. (C) Melting ice needs heat yet only changes the state of the same water, which can be frozen back — a heat-driven physical change that refutes the rule that needing heat makes a change chemical. Caramelising sugar forms new substances (chemical), heating limestone gives off a gas and leaves a new solid (chemical), and burning magnesium makes magnesium oxide (chemical). All three of those are genuinely chemical, so only the melting of ice shows that needing heat does not make a change chemical.

34. (B) Air contains a little carbon dioxide, which reacts even slowly with limewater to form insoluble calcium carbonate, appearing as a faint white skin — the same chemical reaction as the limewater test, just slower. Evaporation would leave behind dissolved lime, not a fresh white insoluble layer formed by reaction, and would not make a distinct skin in a corked bottle. Limewater does not freeze at ordinary room temperature. Oxygen dissolving does not bleach or whiten limewater; the white solid is the carbonate product of reaction with carbon dioxide.

35. (B) Photosynthesis builds glucose and releases oxygen from carbon dioxide and water — these products have different properties from the reactants, and forming new substances is exactly what makes a change chemical, regardless of any shade change in the leaf. Merely absorbing sunlight is not in itself a chemical change. Water turning to vapour is a physical state change, not the reason photosynthesis is chemical. Placing the plant in the dark does not 'reverse' photosynthesis (respiration is a separate reaction), and reversibility is not the test for being chemical.

36. (D) Gaining or losing magnetism changes only a physical property of the needle; it stays the same steel with no new substance, so both magnetising and de-magnetising are physical changes. Magnetic 'energy' is not a substance, so neither step is chemical. There is no new substance in either, so the mixed-label option is wrong. Brief heating to de-magnetise is not the same as rusting, which needs moisture and air over time and forms a new oxide; here the needle stays steel.

37. (B) Dissolving salt in either cup is a physical change: the salt spreads through the water but stays salt and can be recovered by evaporation, and no new substance or visible reaction appears in either cup. Two dissolved substances do not 'always react'; here nothing reacts. The salt is not destroyed — it can be recovered by evaporating the water.

The presence of colour from copper sulphate does not make the dissolving chemical; the salt simply dissolves in both, so both are physical.

38. (A) Coastal air is humid and salty; moisture is one of the two essentials for rusting, and dissolved salt speeds the process, so a railing near the sea rusts faster. Heat alone does not rust iron — without moisture and oxygen there is no rust. Carbon dioxide is not the gas responsible for rusting; oxygen is. The railing is not a 'weaker iron that rusts without air or moisture'; rusting always needs oxygen and water, and the sea simply supplies more moisture and salt.

39. (A) A chemical change is defined by forming a new substance, and chemical changes can either release or absorb heat. The first change (heat out, new substance) and the second (heat in, new substance) are both chemical, while the third (heat out, no new substance) is physical, like dissolving certain solids that warm the water. So exactly two are chemical. It is not 'only releasing heat' that marks chemistry, not 'any heat involvement', and certainly not that heat always means physical.

40. (B) Hammering the red-hot rod only changes its shape, and a red glow is just very hot iron radiating light; in both events the metal stays iron with no new substance, so both are physical. Heating iron until it glows does not turn it into a new substance, so 'both chemical' is wrong. The glow is incandescence, not proof of a reaction, so the mixed label is wrong. Quick shaping and quenching is not rusting (which needs time, moisture and air) nor crystallisation (which means crystals coming out of a solution).

41. (A) Crystallisation separates a dissolved solid out as crystals of the very same substance, so it is physical, whereas rusting combines iron with oxygen and water to make a new substance, hydrated iron oxide, so it is chemical — the two differ in whether a new substance forms. They are not both chemical: crystallisation makes no new substance. They are not both physical: rusting does. The last option reverses the truth, since crystallisation is physical and rusting is chemical.

42. (B) To isolate oxygen while keeping water present, both set-ups must be wet and differ only in air access: a nail in water open to air (oxygen available) versus a nail in boiled water sealed under oil (air expelled) does exactly that, and only the open one rusts, showing oxygen is needed. The dry-versus-moist pair tests moisture, not oxygen, and is not kept wet in both. Salty versus plain water tests salt, with oxygen present in both. Painted versus bare changes the barrier, not cleanly the oxygen, so it does not isolate oxygen.

43. (B) Adding water only disperses the same colouring substance through a larger volume, so the colour looks paler; the dye is unchanged and could be recovered by evaporating the water, making this a physical change. A colour change is not automatic proof of chemistry — here it is just dilution. Water does not react with the dye to weaken it;

it merely dilutes. It is decidable: the recoverability test shows no new substance, so it is physical, not undecidable.

44. (D) Cooking the egg is chemical because new substances form from the proteins, but a smashed shell is also irreversible while staying the same shell material — a physical change — which shows irreversibility by itself does not prove chemistry. So the cooking verdict is correct but the reasoning (irreversible therefore chemical) is flawed. Irreversibility does not guarantee chemistry. The cooking is genuinely chemical, so 'both physical' is wrong, and cooking does form new substances, so 'neither forms a new substance' is wrong.

45. (A) Failing to recover the original by any physical means is strong evidence that a new substance formed, which is the defining test for a chemical change. A temperature rise alone is not proof, since some physical changes (such as dissolving certain solids) also release heat. The absence of a colour change does not prove chemistry — many chemical changes show no colour change and many physical ones do. The absence of a gas likewise proves nothing about new substances. So only the recoverability result is decisive here.

46. (C) Melting ice is a state change of the same water (physical), while bread going mouldy involves microbes producing new substances and a new smell (chemical), so the distinguishing test is whether new substances appear. Being 'left out in the air' does not make a change physical or chemical by itself, so 'both physical' is wrong. Warmth alone does not make a change chemical, so 'both chemical' is wrong. Absorbing heat does not make melting chemical, and mould forms new substances, so the last option is reversed.

47. (B) When copper is heated in air, its surface combines with oxygen to form a black copper-oxygen compound — a new substance, so it is a chemical change, which is why fresh copper lies beneath the black layer. Soot is loose flame deposit, but here a layer forms by reaction with oxygen on clean copper even in a clean flame. Hot copper glows but does not stay black after cooling unless a new substance formed. The wire was not melted and re-frozen; the black is a chemical product, not a change of state.

48. (D) Energy can flow either way in both kinds of change: melting absorbs heat and freezing releases it (physical), while burning releases heat and some reactions absorb it (chemical), so energy flow by itself does not reveal whether a change is physical or chemical. It is false that only chemical changes release energy (freezing releases heat). It is false that chemical changes always release energy or that physical changes never absorb it. Sorting changes by energy direction does not match the physical/chemical divide, so the last option is wrong.

49. (A) As the water evaporates the dissolved sugar comes out as crystals of the very same sugar, which is crystallisation, a physical change. Where the sugar dried and overheated at the edge it browned into a new substance, a chemical change. So one event is physical and the other chemical. It is not 'both physical', because the browning makes a new substance. It is not 'both chemical', because crystals of the same sugar are not a new substance. The last option reverses the labels — crystallising is the physical one and browning the chemical one.

50. (B) Melting solder and letting it solidify are changes of state of the same metal, which keeps its identity and can be re-melted unchanged, so they are physical changes; being a metal does not turn melting into a chemical change. Heat does not change the solder into a new substance, so 'melting a metal is chemical' is wrong. Cooling back to a lump gives the same metal, so it is not a new solid. Briefly melting on a lid is not rusting, which needs moisture and time and forms a new oxide.

51. (D) A burning candle famously shows both: solid wax melting to liquid is a physical state change, while the wax vapour combining with oxygen at the flame to form new gases and soot is a chemical change. The other options pair a physical event with something that is also physical: the drink getting colder is just heat moving (physical), water turning salty is still just dissolved salt (physical), and a nail warming from bending is heat from friction, not a new substance. Only the candle has a true chemical change alongside the physical one.

52. (D) Warmth (heat released), a precipitate, a new smell, an evolved gas, and the inability to recover the original by a physical means are all recognised clues that point to a chemical change here — that is five. While heat or a gas alone might be ambiguous, in this combined set each item is acting as a chemical-change clue, and the recovery failure clinches that a new substance formed. So the count is not three, two, or four; all five observations are clues in this case.

53. (D) Crystallisation purifies physically: copper sulphate dissolves and is separated from insoluble dirt by filtering, then re-forms as crystals on cooling, while soluble impurities remain dissolved in the mother liquor — so the crystals are the same copper sulphate but cleaner. The impurities are not chemically converted into copper sulphate. Heating to dissolve does not burn impurities away. The crystals are the same compound as the starting copper sulphate, not a brand-new substance, so the purity comes from separation, not from making something new.

54. (A) Intact coatings prevent rusting: grease (W) blocks air and moisture, zinc (Z) both blocks and sacrificially protects, and paint (X) blocks air and moisture, while the bare chain (Y) in a humid shed has both oxygen and moisture and rusts the most. So the bare chain is worst and the protected items are far better. Grease does not attract moisture to speed rusting; it repels it. Iron does not rust equally regardless of storage — coatings make a large difference. Zinc protects iron rather than rusting faster than bare iron, so Z is not the most rusted.

55. (B) Bina is correct: the defining feature of a chemical change is the formation of a new substance, and that can occur even when the visible clues (colour, gas, flame, precipitate) are faint or absent at the moment of observation. Aman is wrong that no visible clue means no new substance — clues are helpful signs, not the definition. Chemical changes are not defined by producing a visible signal. A change with no obvious clue is not automatically physical; one must test whether a new substance formed, so 'neither' is wrong.

56. (A) Burning iron wire in a flame and rusting iron in damp air both have iron combining with oxygen to form new iron-oxygen substances, so both are chemical changes — one rapid, one slow. Speed does not decide the type: a slow change can be chemical, so 'slow changes are physical' is wrong. Both form new substances, so 'both physical' is wrong. Fiery burning is not physical merely because sparks fly; the sparks are bits of iron reacting with oxygen, so the last option misclassifies the burning.

57. (C) Dissolving salt and recovering it by evaporation is reversible and keeps the same substance throughout, supporting that this reversible change is physical. Burning a candle cannot be reversed by collecting smoke — it is chemical and irreversible. Cooling a cooked egg does not give back the runny egg — cooking is chemical and irreversible. Drying a rusted nail does not turn rust back into shiny iron — rusting is chemical and not undone by drying. Only the salt example is a reversible physical change.

58. (C) Paint protects only as a barrier; a chip lets air and moisture reach the bare iron, which rusts, and because rust is bulky and flaky it lifts the surrounding paint, baring more iron that then rusts too — so the spot spreads. The paint does not itself turn into rust; only the iron does. There is no 'paint solvent' attacking the iron through the chip. Paint slows rusting compared with bare iron; the spreading spot is due to the breach, not because paint speeds rusting, so the last option is wrong.

59. (C) The boiled-water-sealed nail had water but no dissolved air and did not rust, which shows water by itself is not enough — air (oxygen) is also required. The open-water nail rusted because it had both water and air. The dry-air nail had air but no moisture and did not rust, so air alone is not enough either. Both water and air are needed together. 'Neither needed' is contradicted by the open-water nail rusting precisely when both were present, not from being disturbed.

60. (D) The false idea is that producing a gas always means a chemical change: boiling water produces steam (a gas) and opening a fizzy drink lets dissolved gas escape, both physical, so gas evolution is only sometimes a chemical clue. The other three are true: a chemical change forms new substances while a physical change does not; freezing is a physical change that releases heat; and crystallisation yields pure crystals of the same substance and is physical.